

**FLUID MECHANICS I**  
**PROBLEM 1 (10 POINTS).**

**24-01-2020**  
**Time: 60 minutes**

NAME:.....

GRADE:.....

A incompressible fluid has constant density  $\rho$ , specific heat  $c$  and thermal conductivity  $k$ , while the viscosity  $\mu$  is a function of temperature according to

$$\mu(T) = \mu_0 \left( \frac{T}{T_0} \right)^{3/2} \quad (1)$$

The fluid moves between two infinitely long stagnant walls separated a distance  $h$ . The flow is two-dimensional in the plane  $(x, y)$ , where  $x$  is the coordinate parallel to the walls and  $y$  is perpendicular to the walls. The lower wall temperature is  $T_0$  and the upper wall temperature is  $2T_0$ . Because of the geometry, the velocity and the temperature are functions only of the distance to the walls.

1. Find the vertical component  $v$  of the velocity field.
2. Determine the temperature field. Neglect the viscous dissipation.
3. write the viscous stress tensor  $\bar{\tau}'$ , as a function of the distance  $y$  to the lower wall, the velocity  $u(y)$  and its derivatives.
4. Prove that the pressure gradient  $\partial P / \partial x$  is constant.
5. Find the horizontal component  $u(y)$  of the velocity, assuming the pressure gradient  $\partial P / \partial x$  is known.
6. Formulate the problem to be solved if the viscous dissipation is not neglected.

**HINT:**

$$\int \frac{\xi + C}{(1 + \xi)^{3/2}} d\xi = \frac{4 + 2\xi - 2C}{\sqrt{1 + \xi}}$$

**Note:** The momentum equations with variable viscosity  $\mu(T)$  can be written, in cartesian coordinates as:

$$\begin{aligned} \rho \frac{\partial v_x}{\partial t} + \rho v_x \frac{\partial v_x}{\partial x} + \rho v_y \frac{\partial v_x}{\partial y} &= -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \rho f_{m,x} \\ \rho \frac{\partial v_y}{\partial t} + \rho v_x \frac{\partial v_y}{\partial x} + \rho v_y \frac{\partial v_y}{\partial y} &= -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho f_{m,y} \end{aligned}$$

①

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow v = \text{const.}$  B.C.  $y=0: v=0$   
 $y=h: v=0 \Rightarrow \boxed{v=0}$



②

$\rho u \frac{\partial T}{\partial x} + \rho v \frac{\partial T}{\partial y} = k \frac{\partial^2 T}{\partial x^2} + k \frac{\partial^2 T}{\partial y^2} + \rho \phi - \rho \phi \cdot \vec{v}$   
 $y=0: T=T_0$   
 $y=h: T=2T_0$   
 $\Rightarrow \boxed{T = T_0 \left(1 + \frac{y}{h}\right)}$

③

SINCE  $\nabla \cdot \vec{v} = 0 \Rightarrow \vec{v} = \mu \begin{pmatrix} 2 \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} & 2 \frac{\partial v}{\partial y} \end{pmatrix} = \mu_0 \left(1 + \frac{y}{h}\right)^{3/2} \begin{pmatrix} 0 & \frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial y} & 0 \end{pmatrix}$

④

MOMENTUM - x:  $\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \tau_{xx} + \frac{\partial}{\partial y} \tau_{xy}$   
 $0 = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial y} \left( \mu_0 \left(1 + \frac{y}{h}\right)^{3/2} \frac{\partial u}{\partial y} \right)$   
 MOMENTUM - y:  $\rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x} \tau_{xy} + \frac{\partial}{\partial y} \tau_{yy}$   
 $\Rightarrow \frac{\partial p}{\partial y} = 0 \Rightarrow p = p(x) \text{ at most}$   
 SINCE  $\frac{\partial p}{\partial x}$  is function of x and  $\frac{\partial}{\partial y} \left( \mu_0 \left(1 + \frac{y}{h}\right)^{3/2} \frac{\partial u}{\partial y} \right) = f(y) \Rightarrow$   
 ONLY POSSIBLE IF  $\frac{\partial p}{\partial x} = \text{const.}$

⑤

$0 = -\frac{\partial p}{\partial x} + \mu_0 \frac{\partial}{\partial y} \left[ \left(1 + \frac{y}{h}\right)^{3/2} \frac{\partial u}{\partial y} \right] \Rightarrow \left(1 + \frac{y}{h}\right)^{3/2} \frac{du}{dy} = \frac{h}{\mu_0} \frac{\partial p}{\partial x} \left(\frac{y}{h} + 1\right)$

$du = \frac{h^2}{\mu_0} \frac{\partial p}{\partial x} \frac{\left(\frac{y}{h} + 1\right)}{\left(1 + \frac{y}{h}\right)^{3/2}} d\frac{y}{h} \rightarrow u - 0 = \int_0^{\xi} \frac{\xi + 1}{(1 + \xi)^{3/2}} d\xi \frac{h^2}{\mu_0} \frac{\partial p}{\partial x}$   
 $\xi = \frac{y}{h}$

$\boxed{u = \frac{h^2}{\mu_0} \frac{\partial p}{\partial x} \left[ \frac{4 + 2\xi - 2C}{(1 + \xi)^{1/2}} \right]_0^{\xi} = \frac{h^2}{\mu_0} \frac{\partial p}{\partial x} \left[ \frac{4 + 2\xi - 2C}{(1 + \xi)^{1/2}} - \frac{4 - 2C}{1} \right]}$

C is determined by  $u(\xi=1)=0$   $\frac{4 + 2 - 2C}{\sqrt{2}} = 4 - 2C \rightarrow \frac{3 - C}{\sqrt{2}} = 2 - C$

$2\sqrt{2} - 3 = (\sqrt{2} - 1)C$

$\boxed{C = \frac{2\sqrt{2} - 3}{\sqrt{2} - 1}}$

⑥ IF  $\phi_1 = \mu \left( \frac{du}{dy} \right)^2$  NOT NEGLIGIBLE:

$k \frac{\partial^2 T}{\partial y^2} + \mu \left( \frac{du}{dy} \right)^2 = 0$   
 $y=0: T=T_0; u=0$   
 $y=h: T=2T_0; u=0$   
 $-\frac{dp}{dx} + \mu_0 \frac{d}{dy} \left( \left(1 + \frac{y}{h}\right)^{3/2} \frac{du}{dy} \right) = 0$

A MORE COMPLEX SYSTEM OF EQUATIONS MUST BE SOLVED.